

1. Design often requires studying human behavior. Why is that? Mark all correct options.[cite: 1] ☒ A designer might be interested in what particular functions an artefact should satisfy by studying potential users' demands.[cite: 1] ☐ A designer might want to find out some previously hidden causal relationship in the artefact itself in order to achieve a better understanding of the artefact.[cite: 1] ☒ A designer might want to perform a design validation to check whether the artefact fulfils some functional requirements.[cite: 1] ☐ A designer might be interested in the artefact's material, and whether the artefact could be constructed from some other material which has a certain virtue.[cite: 1]

2. In which ways do social science methods differ from natural science methods? Mark all correct options.[cite: 2] ☐ Validity of results is of no concern.[cite: 2] ☐ They produce qualitative data.[cite: 2] ☒ The focus is on human behaviour.[cite: 2] ☐ There are no laboratory experiments.[cite: 2] ☒ Interpretation of data typically involves attributions of mental states.[cite: 2]

3. What does a belief-desire explanation consist in? Mark all correct options.[cite: 3] ☒ The explanandum is explained through an explanans which refers to an individual's beliefs and desires.[cite: 3] ☐ The explanans includes beliefs and desires of an individual which together inductively infer the action to be explained - the explanandum.[cite: 3] ☒ The explanandum is deductively inferred from an individual's beliefs and desires.[cite: 3] ☐ A belief-desire explanation of an individual's action explains why this individual acted the way she did only by reference to her desires.[cite: 3]

4. What do you do when you adopt the intentional stance? Mark all correct options.[cite: 4] ☒ You ascribe a sort of rationality to the object – that is, rationality in the sense that the object has beliefs about how to reach its goals.[cite: 4] ☒ You attribute mental states to the object that you adopt the intentional stance towards.[cite: 4] ☒ You try to understand what the object's goal(s) is in order to understand how the object believes that it will reach this goal.[cite: 4] ☒ You eventually come to a conclusion about what the object ought to do relative the beliefs and desires you have ascribed to the object.[cite: 4]

5. Which of the following are correct ways to understand what the methodological problem of interpretation is about? In other words, what is at stake according to this problem? Mark all correct options.[cite: 5] ☒ The methodological problem of interpretation points to the fact that the field of design frequently necessitate the study of human behavior? **Select all valid reasons.** ☒ To investigate the specific functional requirements an artifact must meet by evaluating the needs of potential users. ☐ To uncover previously unseen causal links within the artifact itself, thereby gaining a better mechanical understanding of it. ☒ To execute design validation, ensuring that the artifact successfully fulfills established functional criteria. ☐ To analyze the artifact's construction material and assess if a different material with advantageous properties could be substituted.

2. In what specific ways do the methodologies of the social sciences diverge from those of the natural sciences? Select all correct answers. ☐ The validity of the research outcomes is not a priority. ☐ They are exclusively responsible for producing qualitative data. ☒ Their primary analytical focus centers on human behavior. ☐ They completely eschew the use of laboratory experiments. ☒ Making sense of the data typically requires attributing mental states to the subjects being studied.

3. What constitutes an explanation based on the belief-desire model? Select all correct statements. ☒ The phenomenon being explained (explanandum) is accounted for by an explanans that directly references the beliefs and desires of an individual. ☐ The explanans contains an individual's beliefs and desires, which combined, allow for an inductive inference of the explanandum (the action). ☒ The action to be explained (explanandum) is deduced logically from the individual's respective beliefs and desires. ☐ This type of explanation solely references an individual's desires to explain why they acted in a particular manner.

4. Which of the following describe what you are doing when adopting the intentional stance? Select all that apply. ☒ You attribute a specific form of rationality to the entity, operating under the assumption that

it holds beliefs regarding how to achieve its aims. ☒ You assign mental states to the target entity you are analyzing. ☒ You attempt to figure out the entity's objectives so you can comprehend how it believes it will accomplish them. ☒ You ultimately determine how the entity should act, given the specific set of beliefs and desires you have ascribed to it.

5. Which statements correctly characterize the stakes involved in the methodological problem of interpretation? Select all correct options. ☐ It emphasizes that even with a framework for modeling behavior, we require better criteria to differentiate good modeling from bad modeling in specific situations.

☒ It highlights the vulnerability that any given interpretation of a behavior might be selected somewhat arbitrarily. ☐ It dictates that any behavior can only be rooted in a single, specific belief and a single desire. ☐ It argues that the fundamental architecture of the belief-desire explanatory model is inherently flawed.

6. True or False: It is impossible to conduct social science research without engaging in interpretation. ☒ True ☐ False

7. What are potential strategies for circumventing the methodological problem of interpretation? Select all correct options. ☒ Whenever feasible, actively reduce the range of potential interpretations. ☐ Ignore social properties entirely and concentrate exclusively on individual-level properties. ☒ When possible, look for observable outcomes through diverse operationalizations to supply additional evidence for attributing a particular mental state. ☒ If practical and logically sound, shift the investigation to focus on stimulus-and-response relationships instead.

8. How can a researcher best guarantee the repeatability of a qualitative observation? Select the correct answer. ☒ By verifying that the interpretive methods applied are intersubjective, meaning the process of interpretation is fully transparent. ☐ By explicitly asserting that another researcher utilizing the exact same methods would invariably observe the identical outcome. ☐ By ensuring that employing a separate operationalization for the feature in question yields adequately comparable results.

9. True or False: It is possible to extract quantitative data from qualitative sources. ☒ True ☐ False

10. Imagine a qualitative observation of participants' emotional attitudes conducted via a survey. Which of the following scenarios would be problematic for this operationalization? Select all correct options. ☒ The individual designing the survey makes the unsubstantiated assumption that all respondents interpret the survey wording in the exact same way. ☐ The survey fails to produce attitude measurements that can be mapped onto an interval or ratio scale. ☐ The emotional attitudes are charted on a measurement scale that does not effectively capture the true magnitude of the participant's attitude (e.g., failing to act as an absolute scale).

11. Which of the following are viable methods for increasing the robustness of a qualitative observation? Select all correct options. ☒ By developing multiple operationalizations for the same property, with the explicit intention of demonstrating convergent validity. ☒ By employing triangulation—utilizing several different observational methods to investigate the exact same property. ☐ By establishing multiple operationalizations for entirely separate properties, with the intention of demonstrating divergent validity. ☐ By ensuring that the final result of the qualitative observation can be successfully reformatted into a quantitative measurement.

12. For what reasons would a researcher choose to utilize a case study methodology? Select all correct options. ☒ Because they can provide foundational data that supports subsequent investigations into highly complex phenomena. ☒ Because case studies excel at establishing cause-and-effect relationships within specific, individual cases, whereas multivariate studies often fall short in this regard. ☐ Because they offer vital data regarding the definitive boundaries of a given demographic population. ☒ Because the findings from a single case study can be used to disprove a theoretical statement that was previously held to be true. ☐ Because case studies inherently possess the capacity to generalize their

findings across much broader populations with ease. ☐ Because experimental artifacts are entirely absent within case study frameworks.

13. True or False: The analytical narrative method shares conceptual similarities with the hypothetico-deductive (H-D) approach. In this method, a researcher selects cases, crafts a narrative aligned with the research question, and deduces observable implications from those cases under the assumption that the narrative is accurate. The researcher then tests these implications to either reinforce or diminish confidence in the narrative. ☒ True ☐ False

14. True or False: A case study is fundamentally never superior to a multivariate study; researchers only rely on them when they lack the necessary data to perform multivariate analysis. ☐ True ☒ False

15. Regarding prevalent misconceptions about case studies, which of the following statements represent true claims about their capabilities? Select all correct options. ☒ In contrast to multivariate analyses, case studies have the unique ability to pinpoint causal mechanisms within specific, individual situations. ☐ Even if case study results are highly valid, they are strictly incapable of being generalized beyond their immediate context. ☐ Case studies completely lack the capacity to support methodologically sound inferences. ☒ Case studies can be highly effective in generating a deeper and more comprehensive explanatory understanding. ☐ It is impossible to incorporate quantitative techniques within a case study methodology.